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JEE MAIN PHYSICS DPP: SPEED & VELOCITY

Topic: Average/Instantaneous Speed & Velocity, Uniform & Non-Uniform Motion | Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** A particle moves along a straight line such that its displacement x at time t is given by $x = 2t^3 - 9t^2 + 12t - 4$ m. The instantaneous velocity of the particle is zero at times:
- $t = 1$ s and $t = 2$ s
 - $t = 2$ s and $t = 3$ s
 - $t = 1$ s and $t = 3$ s
 - $t = 0.5$ s and $t = 2$ s
- Q2.** A bird flies due north with speed 20 m/s for 15 s. It then rests for 5 s and flies due south with speed 25 m/s for 10 s. The average velocity and average speed of the bird over the 30 s interval are:
- $\frac{5}{3}$ m/s (North), $\frac{55}{3}$ m/s
 - 5 m/s (North), 22.5 m/s
 - $\frac{5}{3}$ m/s (South), $\frac{55}{3}$ m/s
 - 0 m/s, 20 m/s
- Q3.** The velocity of a particle moving along the x -axis is given by $v(t) = v_0 e^{-\lambda t}$, where v_0 and λ are positive constants. The total distance covered by the particle in the time interval $t = 0$ to $t \rightarrow \infty$ is:
- $\frac{v_0}{\lambda}$
 - $\frac{v_0}{\lambda^2}$
 - λv_0
 - ∞
- Q4.** A car travels along a straight road. It travels the first half of the total distance with speed v_0 . The remaining half distance is covered with speed v_1 for half the time and with speed v_2 for the other half time. The mean speed of the car is:
- $\frac{2v_0(v_1 + v_2)}{2v_0 + v_1 + v_2}$
 - $\frac{v_0(v_1 + v_2)}{v_0 + v_1 + v_2}$
 - $\frac{2v_0(v_1 + v_2)}{v_0 + 2(v_1 + v_2)}$
 - $\frac{v_0 + v_1 + v_2}{3}$
- Q5.** The displacement x of a particle moving in one dimension varies with time as $t = \alpha x^2 + \beta x$, where α and β are positive constants. The relation between instantaneous acceleration a and velocity v is:
- $a = -2\alpha v^3$
 - $a = 2\alpha v^3$
 - $a = -2\beta v^2$
 - $a = -2\alpha v^2$
- Q6.** A particle moves along a straight line such that its velocity is $v = 4 + 2(s_1 + s_2)^{1/2}$ where s is displacement. If initially $s = 0$ at $t = 0$, the acceleration of the particle is:
- Variable and depends on $\sqrt{s}$
 - Constant and equal to 2 m/s^2
 - Constant and equal to 1 m/s^2
 - Variable and depends on t
- Q7.** An object moves along a path such that its position vector is $\vec{r}(t) = (3t\hat{i} + 4t^2\hat{j})$ m. The magnitude of instantaneous velocity and angle of velocity with the x -axis at $t = 1$ s are:
- $\sqrt{73} \text{ m/s}$, $\tan^{-1}\left(\frac{8}{3}\right)$
 - 11 m/s , $\tan^{-1}\left(\frac{4}{3}\right)$
 - 5 m/s , $\tan^{-1}\left(\frac{3}{8}\right)$
 - $\sqrt{73} \text{ m/s}$, $\tan^{-1}\left(\frac{3}{8}\right)$
- Q8.** A particle starts from rest and moves along the x -axis with non-uniform acceleration $a = 6t - 2 \text{ m/s}^2$. If it starts from $x = 0$ at $t = 0$, its velocity and position at $t = 2$ s are:
- 8 m/s, 4 m
 - 8 m/s, 8 m
 - 12 m/s, 6 m
 - 4 m/s, 4 m
- Q9.** Which of the following conditions represents non-uniform motion with constant speed?
- Motion of a car on a straight highway with cruise control
 - Motion of a particle along a circular track with constant speed
 - Freely falling body under gravity
 - A ball rolling down a smooth inclined plane
- Q10.** The velocity v of a particle moving along a straight line is related to its displacement x by $v^2 = 108 - 9x^2$. The particle executes:
- Uniform motion
 - Uniformly accelerated motion

- (c) Simple harmonic motion with acceleration proportional to $-x$
- (d) Motion with constant non-zero acceleration
- Q11.** A train accelerates from rest at rate α for time t_1 , then moves with constant speed for time t_2 , and finally decelerates to rest at rate β for time t_3 . If the total distance travelled is S , the maximum velocity attained is:
- (a) $\frac{2S}{t_1 + 2t_2 + t_3}$
- (b) $\frac{S}{t_1 + t_2 + t_3}$
- (c) $\frac{2S}{t_1 + t_2 + t_3}$
- (d) $\frac{S}{2t_1 + t_2 + 2t_3}$
- Q12.** A body moves along a straight line such that its velocity varies with displacement as $v = v_0 - kx$, where v_0 and k are positive constants. The time taken by the body to come to rest is:
- (a) $\frac{1}{k}$
- (b) $\frac{v_0}{k}$
- (c) ∞
- (d) $\frac{\ln 2}{k}$
- Q13.** The acceleration of a particle is given by $a = 4t - 30$, where a is in m/s^2 and t in seconds. If velocity $v = 5 \text{ m/s}$ at $t = 0$, the velocity of the particle when acceleration is zero is:
- (a) -107.5 m/s
- (b) -112.5 m/s
- (c) -117.5 m/s
- (d) -100.0 m/s
- Q14.** A particle moves along a straight line according to $x(t) = a \cos(\omega t)$. The average speed of the particle in the time interval $t = 0$ to $t = \frac{\pi}{\omega}$ is:
- (a) $\frac{2a\omega}{\pi}$
- (b) $\frac{a\omega}{\pi}$
- (c) 0
- (d) $\frac{4a\omega}{\pi}$
- Q15.** The speed-time graph of a particle is a triangle of base T (time axis) and peak height $v_{\max}$. The ratio of average speed to maximum speed is:
- (a) $\frac{1}{2}$
- (b) $\frac{1}{3}$
- (c) $\frac{2}{3}$
- (d) $\frac{1}{4}$
- Q16.** A particle moves along the x -axis with an acceleration $a = -bv$, where v is velocity and b is a constant. At $t = 0$, $v = v_0$ and $x = 0$. The maximum displacement of the particle is:
- (a) $\frac{v_0}{b}$
- (b) $\frac{v_0^2}{2b}$
- (c) $\frac{v_0}{2b}$
- (d) $\frac{2v_0}{b}$
- Q17.** If a particle moves such that its position is $x(t) = 4t - 2t^2 \text{ m}$, the average velocity and average speed in the interval $t = 0$ to $t = 2 \text{ s}$ are:
- (a) $0 \text{ m/s}, 2 \text{ m/s}$
- (b) $0 \text{ m/s}, 0 \text{ m/s}$
- (c) $2 \text{ m/s}, 2 \text{ m/s}$
- (d) $0 \text{ m/s}, 4 \text{ m/s}$
- Q18.** A particle moves in a straight line with deceleration proportional to displacement: $a = -\kappa x$. The decrease in kinetic energy of the particle in moving from x_1 to x_2 is proportional to:
- (a) $x_2^2 - x_1^2$
- (b) $x_2 - x_1$
- (c) $x_2^3 - x_1^3$
- (d) $\sqrt{x_2} - \sqrt{x_1}$
- Q19.** A swimmer crosses a river of width d flowing with velocity v_r . If the speed of the swimmer relative to still water is v_s ($v_s > v_r$), the shortest possible time to cross the river is:
- (a) $\frac{d}{v_s}$
- (b) $\frac{d}{\sqrt{v_s^2 - v_r^2}}$
- (c) $\frac{d}{v_s + v_r}$
- (d) $\frac{d}{v_r}$
- Q20.** A car travels from point A to point B at speed v_1 and immediately returns from B to A at speed v_2 . The magnitude of average velocity for the round trip is:
- (a) 0
- (b) $\frac{2v_1v_2}{v_1 + v_2}$
- (c) $\frac{v_1 + v_2}{2}$
- (d) $\sqrt{v_1v_2}$

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(a)	6	(b)	11	(a)	16	(a)
2	(a)	7	(a)	12	(c)	17	(a)
3	(a)	8	(a)	13	(a)	18	(a)
4	(a)	9	(b)	14	(a)	19	(a)
5	(a)	10	(c)	15	(a)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (a)

$$x(t) = 2t^3 - 9t^2 + 12t - 4.$$

Instantaneous velocity:

$$v(t) = \frac{dx}{dt} = 6t^2 - 18t + 12 = 6(t^2 - 3t + 2)$$

$$\text{Setting } v(t) = 0 \Rightarrow 6(t-1)(t-2) = 0 \Rightarrow t = 1 \text{ s and } t = 2 \text{ s.}$$

Sol 2. Correct Option: (a)

• Displacement: $\vec{s}_1 = 20 \times 15 = +300 \text{ m (North)}$.
 $\vec{s}_2 = 0 \text{ m}$.

$\vec{s}_3 = 25 \times 10 = -250 \text{ m (South)}$.

Net displacement = $300 - 250 = +50 \text{ m (North)}$.

Average velocity = $\frac{50 \text{ m}}{30 \text{ s}} = \frac{5}{3} \text{ m/s (North)}$.

• Total distance = $300 + 0 + 250 = 550 \text{ m}$.

Average speed = $\frac{550 \text{ m}}{30 \text{ s}} = \frac{55}{3} \text{ m/s}$.

Sol 3. Correct Option: (a)

Since $v(t) > 0$ for all t , distance equals displacement:

$$s = \int_0^\infty v(t) dt = \int_0^\infty v_0 e^{-\lambda t} dt$$

$$= v_0 \left[-\frac{e^{-\lambda t}}{\lambda} \right]_0^\infty = \frac{v_0}{\lambda} (0 - (-1)) = \frac{v_0}{\lambda}$$

Sol 4. Correct Option: (a)

Let total distance be $2s$.

• Time for first half: $t_1 = \frac{s}{v_0}$.

• For second half distance s , let total time taken be $2t'$. Distance covered: $s = v_1 t' + v_2 t' = (v_1 + v_2)t' \Rightarrow t' = \frac{s}{v_1 + v_2}$.

Time for second half distance = $2t' = \frac{2s}{v_1 + v_2}$.

Average speed:

$$v_{\text{avg}} = \frac{2s}{t_1 + 2t'} = \frac{2s}{\frac{s}{v_0} + \frac{2s}{v_1 + v_2}}$$

$$= \frac{2v_0(v_1 + v_2)}{2v_0 + v_1 + v_2}$$

Sol 5. Correct Option: (a)

$$t = \alpha x^2 + \beta x.$$

Differentiating with respect to t :

$$1 = (2\alpha x + \beta) \frac{dx}{dt} \Rightarrow v = (2\alpha x + \beta)^{-1}$$

Differentiating again with respect to t :

$$a = \frac{dv}{dt} = -(2\alpha x + \beta)^{-2} \left(2\alpha \frac{dx}{dt} \right)$$

$$= -v^2(2\alpha v) = -2\alpha v^3$$

Sol 6. Correct Option: (b)

$$v = 4 + 2s^{1/2}.$$

Differentiating with respect to s :

$$\frac{dv}{ds} = 2 \left(\frac{1}{2} s^{-1/2} \right) = s^{-1/2}$$

Acceleration: For $v = 2\sqrt{s}$, $a = v \frac{dv}{ds} = (2\sqrt{s}) \left(\frac{1}{\sqrt{s}} \right) = 2 \text{ m/s}^2$ (constant).

Sol 7. Correct Option: (a)

$$\vec{r}(t) = 3t\hat{i} + 4t^2\hat{j}.$$

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = 3\hat{i} + 8t\hat{j}.$$

At $t = 1 \text{ s}$, $\vec{v} = 3\hat{i} + 8\hat{j} \text{ m/s}$.

$$|\vec{v}| = \sqrt{3^2 + 8^2} = \sqrt{9 + 64} = \sqrt{73} \text{ m/s}.$$

$$\theta = \tan^{-1} \left(\frac{v_y}{v_x} \right) = \tan^{-1} \left(\frac{8}{3} \right).$$

Sol 8. Correct Option: (a)

$$a = \frac{dv}{dt} = 6t - 2 \Rightarrow v(t) = \int (6t - 2) dt = 3t^2 - 2t + C.$$

Since $v(0) = 0 \Rightarrow C = 0$.

At $t = 2 \text{ s}$: $v(2) = 3(2)^2 - 2(2) = 12 - 4 = 8 \text{ m/s}$.

Position $x(t) = \int (3t^2 - 2t) dt = t^3 - t^2$.

$$x(2) = (2)^3 - (2)^2 = 8 - 4 = 4 \text{ m}.$$

Sol 9. Correct Option: (b)

In uniform circular motion, the speed $|v|$ is constant, but the direction of velocity changes continuously ($\vec{a}_c = \frac{v^2}{r} \hat{r} \neq 0$), representing non-uniform velocity motion at constant speed.

Sol 10. Correct Option: (c)

$$v^2 = 108 - 9x^2.$$

Differentiating with respect to x :

$$2v \frac{dv}{dx} = -18x \implies a = v \frac{dv}{dx} = -9x$$

Since acceleration is directly proportional to displacement and directed opposite to it ($a = -\omega^2 x$), the particle executes SHM.

Sol 11. Correct Option: (a)

From the velocity-time graph (trapezium):

- Base 1 = $t_1 + t_2 + t_3$

- Base 2 = t_2

- Height = $v_{\max}$

$$\text{Area} = S = \frac{1}{2}(t_1 + 2t_2 + t_3)v_{\max}.$$

$$\text{Thus, } v_{\max} = \frac{2S}{t_1 + 2t_2 + t_3}.$$

Sol 12. Correct Option: (c)

$$v = \frac{dx}{dt} = v_0 - kx \implies \frac{dx}{v_0 - kx} = dt.$$

Integrating:

$$-\frac{1}{k} \ln \left(\frac{v_0 - kx}{v_0} \right) = t \implies v = v_0 e^{-kt}$$

$$\text{As } v \rightarrow 0, e^{-kt} \rightarrow 0 \implies t \rightarrow \infty.$$

Sol 13. Correct Option: (a)

$$a = 4t - 30 = 0 \implies t = 7.5 \text{ s.}$$

$$\text{Velocity } v(t) = \int a dt = 2t^2 - 30t + C.$$

$$\text{Given } v(0) = 5 \implies C = 5.$$

$$v(7.5) = 2(7.5)^2 - 30(7.5) + 5 = 112.5 - 225 + 5 = -107.5 \text{ m/s.}$$

Sol 14. Correct Option: (a)

From $t = 0$ to $t = \pi/\omega$, the particle moves from $x = a$ to $x = -a$ without reversing direction.

$$\text{Distance} = |(-a) - a| = 2a.$$

$$\text{Time interval } \Delta t = \frac{\pi}{\omega}.$$

$$\text{Average speed} = \frac{\text{Distance}}{\Delta t} = \frac{2a}{\pi/\omega} = \frac{2a\omega}{\pi}.$$

Sol 15. Correct Option: (a)

$$\text{Total distance} = \text{Area under } v-t \text{ graph} = \frac{1}{2} T v_{\max}.$$

$$\text{Average speed} = \frac{\text{Distance}}{T} = \frac{\frac{1}{2} T v_{\max}}{T} = \frac{1}{2} v_{\max}.$$

$$\text{Ratio: } \frac{v_{\text{avg}}}{v_{\max}} = \frac{1}{2}.$$

Sol 16. Correct Option: (a)

$$a = v \frac{dv}{dx} = -bv \implies dv = -b dx.$$

Integrating from $v = v_0$ ($x = 0$) to $v = 0$ ($x = x_{\max}$):

$$\int_{v_0}^0 dv = -b \int_0^{x_{\max}} dx$$

$$-v_0 = -b x_{\max} \implies x_{\max} = \frac{v_0}{b}$$

Sol 17. Correct Option: (a)

$$x(t) = 4t - 2t^2.$$

- $x(0) = 0 \text{ m}, x(2) = 4(2) - 2(2)^2 = 0 \text{ m.}$

$$\text{Displacement} = 0 \implies \text{Average velocity} = 0 \text{ m/s.}$$

- Velocity $v = 4 - 4t = 0 \implies t = 1 \text{ s.}$

$$x(1) = 4(1) - 2(1)^2 = 2 \text{ m.}$$

$$\text{Distance} = |x(1) - x(0)| + |x(2) - x(1)| = 2 + 2 = 4 \text{ m.}$$

$$\text{Average speed} = \frac{4 \text{ m}}{2 \text{ s}} = 2 \text{ m/s.}$$

Sol 18. Correct Option: (a)

Work-Energy Theorem:

$$W = \Delta K = \int_{x_1}^{x_2} m a dx$$

$$= -m\kappa \int_{x_1}^{x_2} x dx = -\frac{1}{2} m\kappa (x_2^2 - x_1^2)$$

$$\text{Decrease in KE} \propto x_2^2 - x_1^2.$$

Sol 19. Correct Option: (a)

To cross in minimum time, the swimmer must swim directly perpendicular to the flow ($\theta = 90^\circ$ to river bank):

$$t_{\min} = \frac{d}{v_s}$$

Sol 20. Correct Option: (a)

Since the round trip returns the car to its starting point A, the net displacement $\Delta \vec{x} = 0$.

Average velocity:

$$\vec{v}_{\text{avg}} = \frac{\Delta \vec{x}}{\Delta t} = \frac{0}{\Delta t} = 0$$

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